SSD PLAYER
开发环境搭建

REVISION HISTORY

Revision No.	Description	Date
0.1	• Initial release	06/06/2019
	•	
	•	

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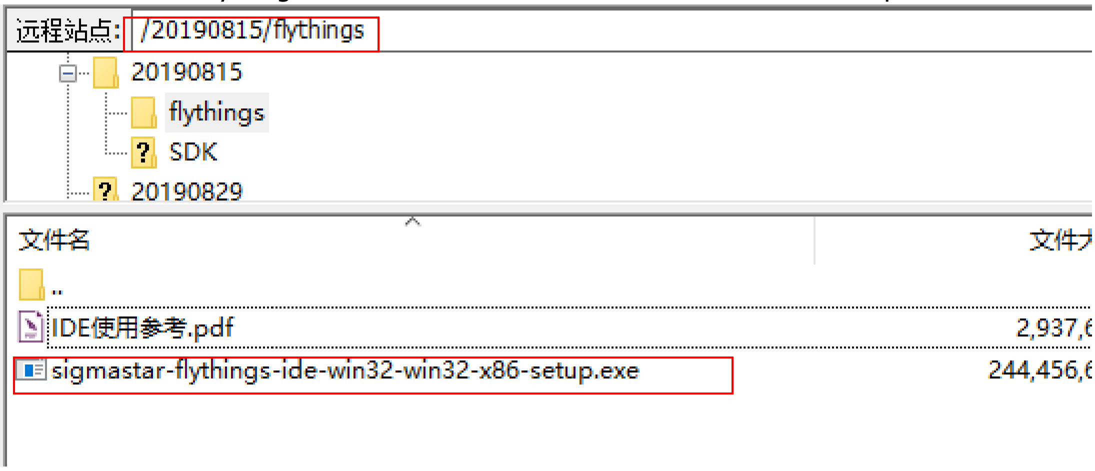
精简版 FLYTHINGS IDE 环境搭建

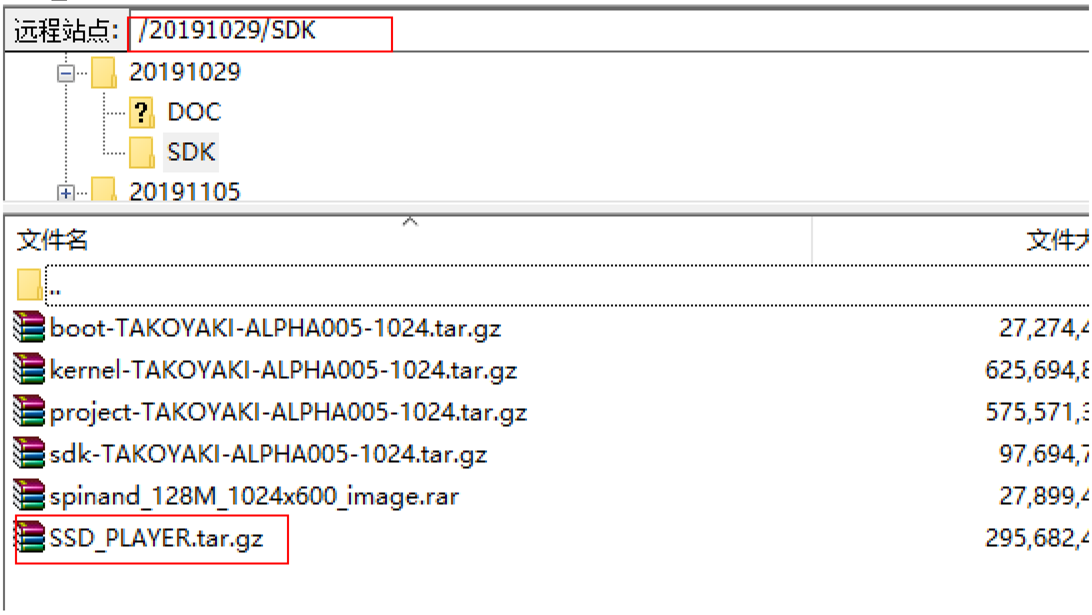
1.1. 安装 flythings 工具

到 FTP 如下路径下载 flythings 工具，并安装，具体使用方法参考:IDE 使用参考.pdf

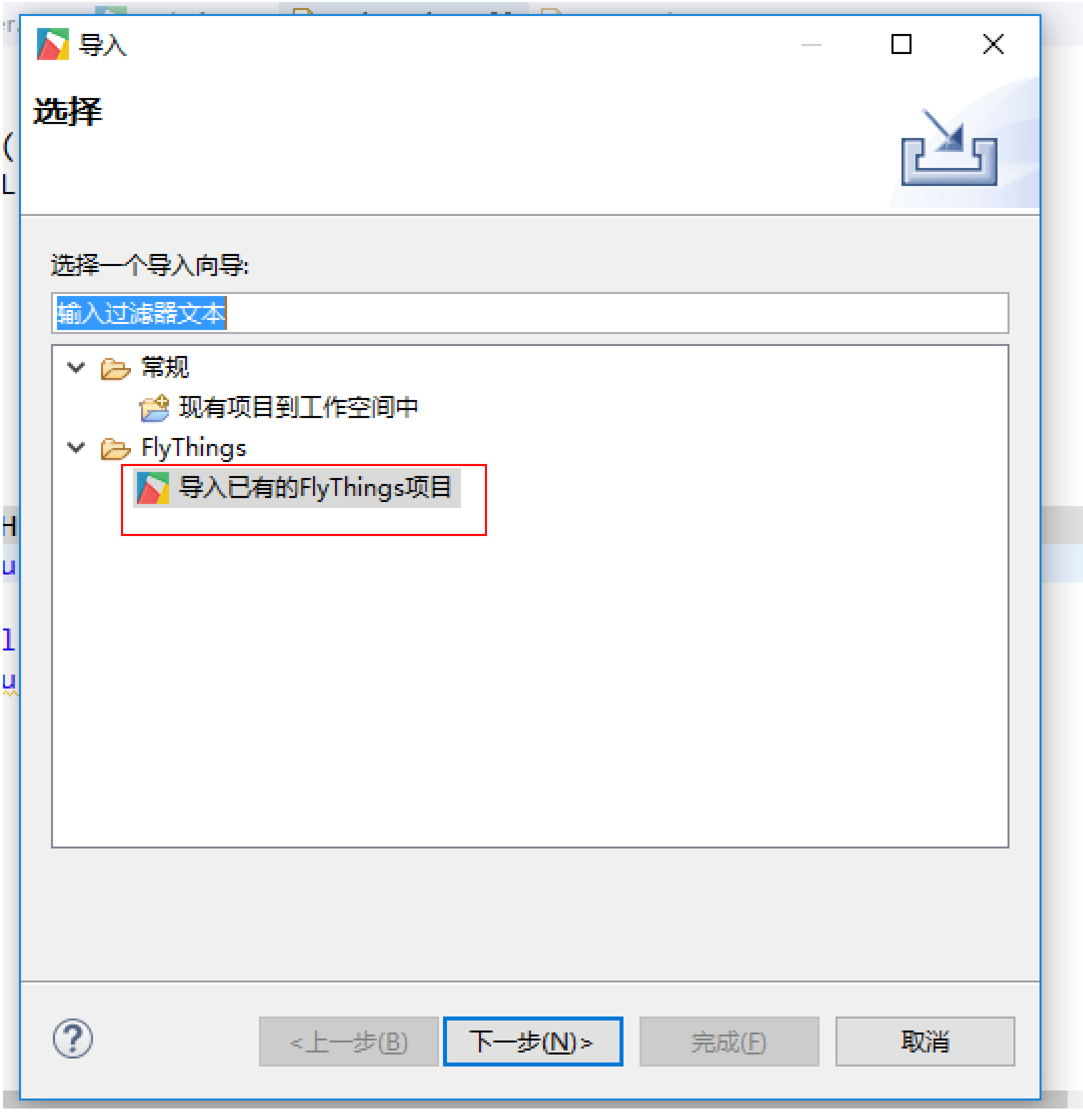


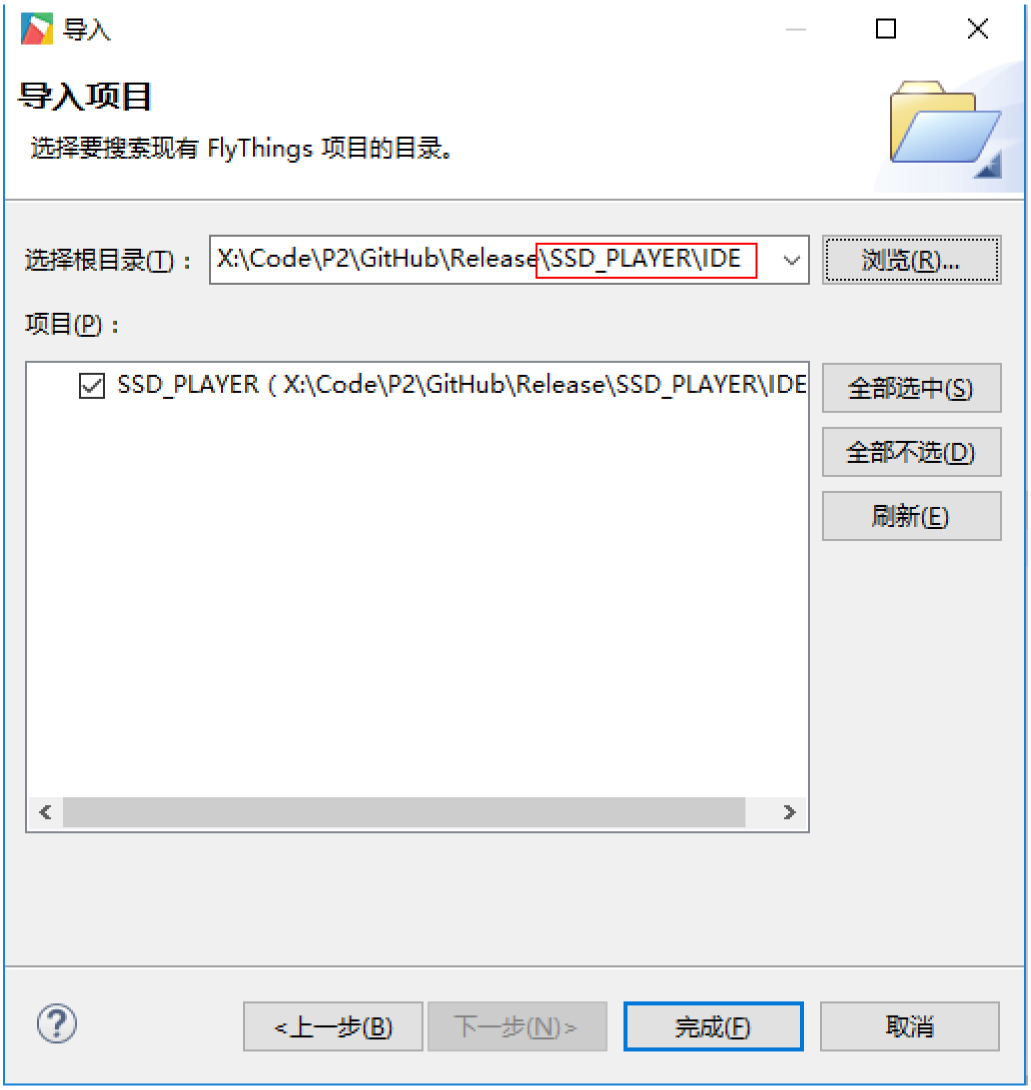
1.2. 2. 导入 SSD_PLAYER 工程

SSD_PLAYER 工程目录放在 FTP 的如下路径：

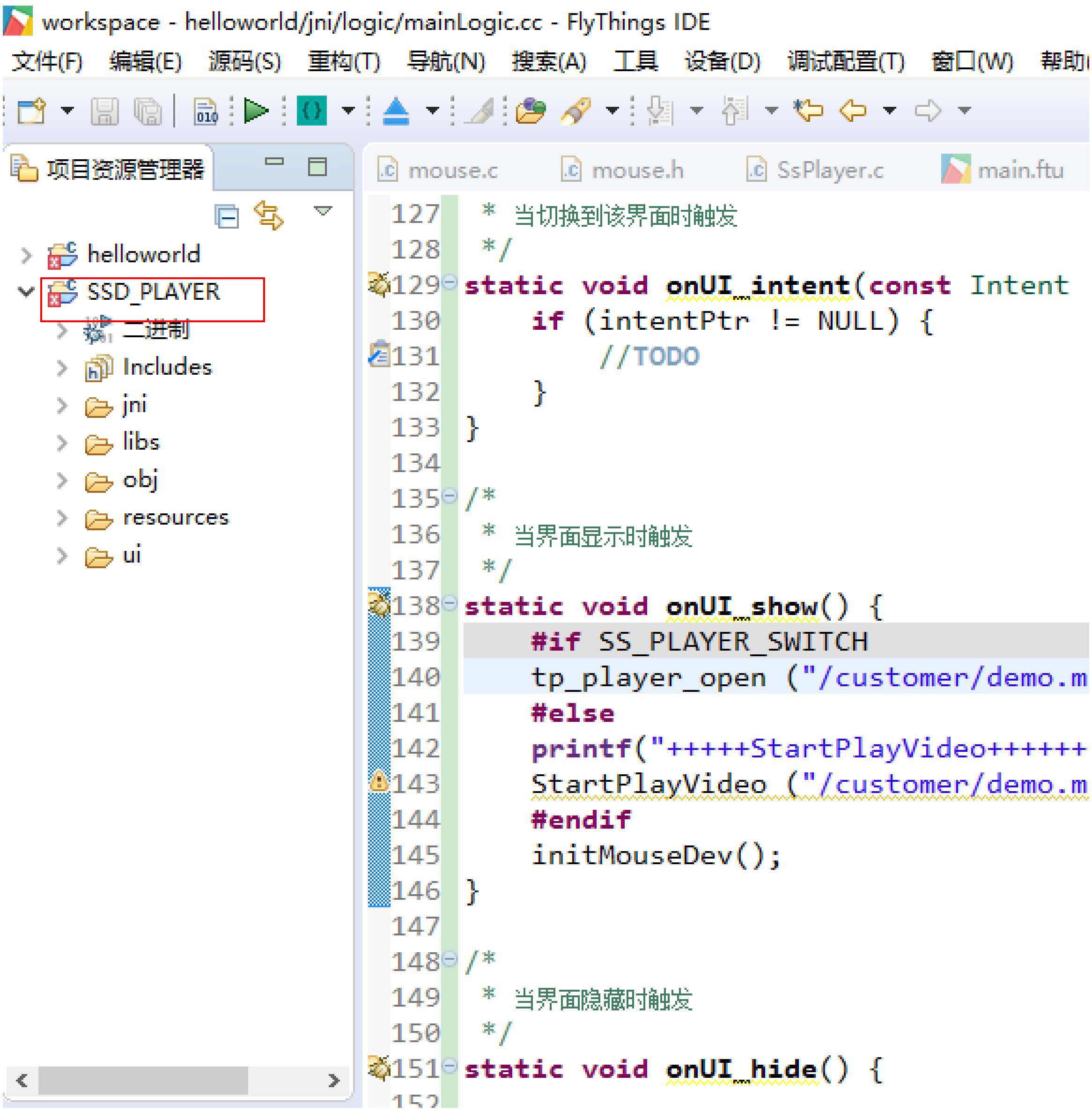


解压后，选择 flythings 文件-导入功能，将 SSD_PLAYER\IDE 导入工程:


导入
选择
选择一个导入向导:
输入过滤器文本
常规
现有项目到工作空间中
FlyThings
导入已有的FlyThings项目
< 上一步(B)
下一步(N) >
完成(F)
取消

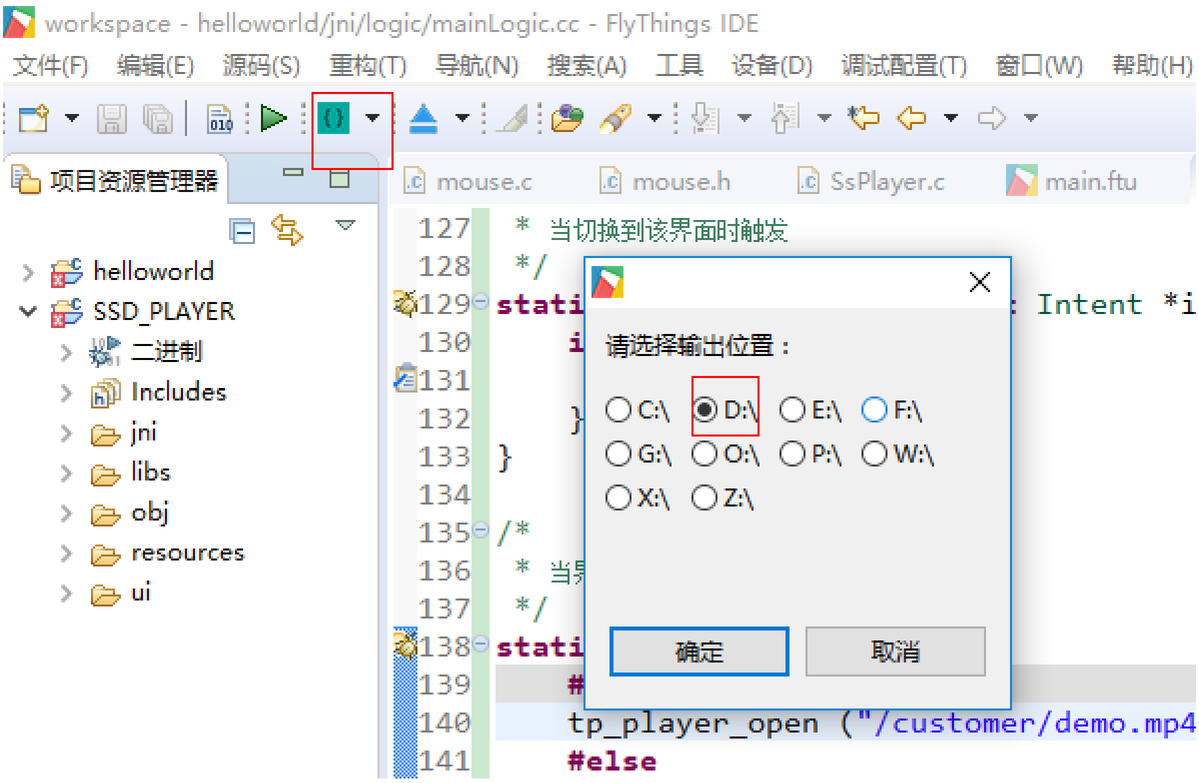


导入后，可以在 flythings 左边的工程栏看到对应的工程：



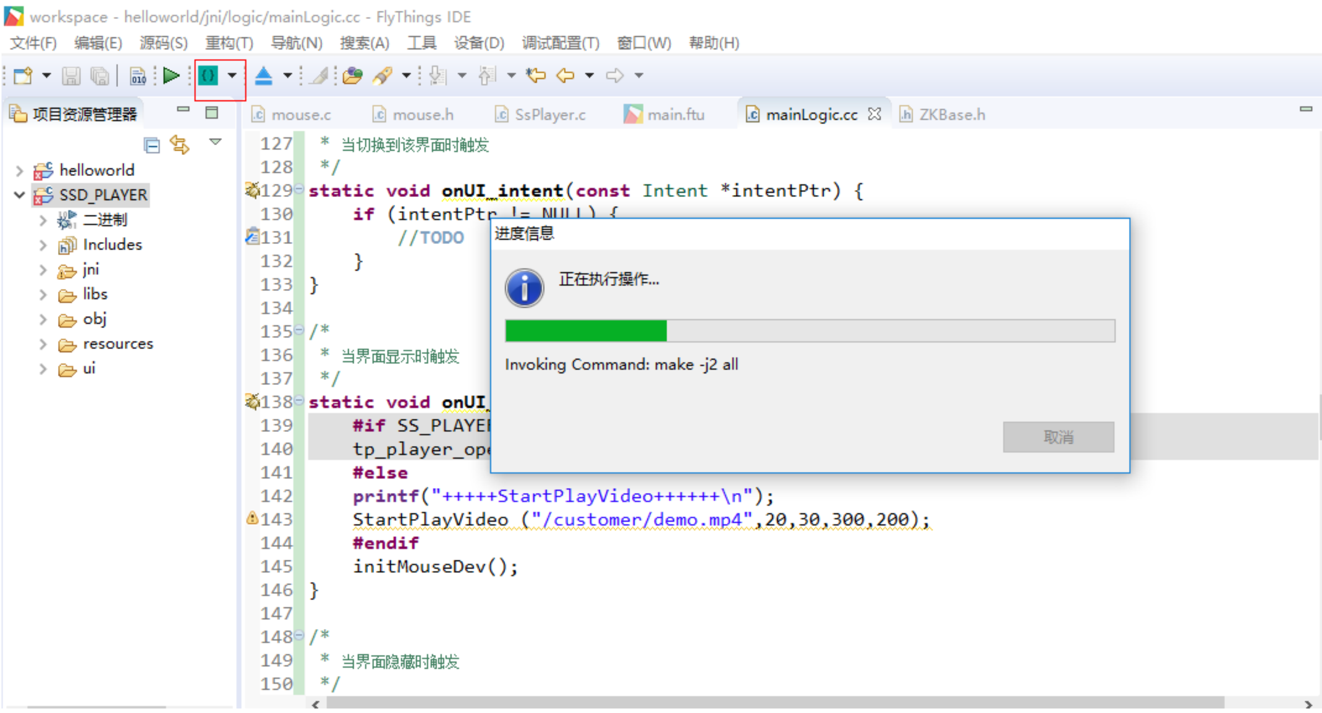
1.3. 设置编译 SSD_PLAYER 生成资源&lib 导出文件路径

设置编译生成的资源和 lib 导出文件路径，目前路径只能存在对应盘符的根目录：



1.4. 编译 SSD_PLAYER 工程

选择左边 SSD_PLAYER 工程，点击右键，选择清空项目，然后点击如下按钮编译 SSD_PLAYER 工程：



编译完成后会在第二步设定的目录下生成 lib 和 res 文件：

此电脑 › 软件 (D:) ›

名称	修改日期	类型	大小
EasyUI.cfg	2019/11/7 12:00	CFG 文件	1 KB
udmadebug.log	2019/10/29 17:44	文本文档	0 KB
zkrestart	2019/10/15 21:21	文件	0 KB
TV_CAM_设备_20190809_110232.mov	2019/8/9 11:02	MOV 文件	6,233 KB
email	2019/11/7 12:08	文件夹	
lib	2019/11/7 12:00	文件夹	
ui	2019/11/6 18:06	文件夹	
font	2019/10/21 12:25	文件夹	
BaiduNetdiskDownload	2019/9/20 10:29	文件夹	

将 lib 和 ui 文件覆盖到 sdk-TAKOYAKI-ALPHA005-1024.tar.gz 解压后的 sdk\verify\application\customer_zk\lib 和 sdk\verify\application\customer_zk\res\ui 目录。

1.5. 编译 project & sdk & kernel & uboot 生成 image

a. 打开 project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk 中的 verify_zk_mini，将 customer_zk_mini app 打包进 customer 分区，并开机自动运行：

```
include $(PROJ_ROOT)/release/customer_tailor/nvr_default.mk
#interface_ai:=disable
#interface_ao:=disable
#interface_gfx:=disable
interface_hdmi:=disable
#interface_divp:=disable
#interface_disp:=disable
#interface_panel:=disable
interface_rgn:=disable
interface_shadow:=disable
interface_uac:=disable
interface_vdf:=disable
interface_vdisp:=disable
#interface_vdec:=disable
interface_venc:=enable
interface_wlan:=enable

misc_fbdev:=enable
verify_zk_mini:=enable
#mhal
#mhal_aio:=disable
mhal_csi:=disable
#mhal_disp:=disable
```

b. 将 FTP 的/20191029/SDK/Release_to_customer.sh 的编译脚本拷贝到 project 同级目录：

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$ ls -la
total 1267324
drwxr-xr-x  6 aaron.feng sigmastar_users      4096 Nov  7 11:52 .
drwxr-xr-x  7 aaron.feng sigmastar_users      4096 Oct 25 17:07 ..
drwxr-xr-x 21 aaron.feng sigmastar_users      4096 Oct 21 18:42 boot
-rw-r--r--  1 aaron.feng sigmastar_users  15800408 Oct 25 17:05 boot-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x 26 aaron.feng sigmastar_users      4096 Nov  6 20:26 kernel
-rw-r--r--  1 aaron.feng sigmastar_users 625576086 Oct 25 17:05 kernel-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x  8 aaron.feng sigmastar_users      4096 Oct 25 17:04 project
-rw-r--r--  1 aaron.feng sigmastar_users 558616668 Oct 25 17:06 project-TAKOYAKI-ALPHA005-1024.tar.gz
-rwxrw-rw-  1 aaron.feng sigmastar_users      2829 Nov  7 10:33 Release_to_customer.sh
drwxr-xr-x  4 aaron.feng sigmastar_users      4096 Oct 25 17:04 sdk
-rw-r--r--  1 aaron.feng sigmastar_users  97694764 Nov  6 20:49 sdk-TAKOYAKI-ALPHA005-1024.tar.gz
```

c. 根据 flash type 和芯片型号运行 Release_to_customer.sh 编译 project 生成 image 文件：

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$ ./Release_to_customer.sh -p ssd202 -f nand
chose nand
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  SHIPPED scripts/kconfig/zconf.tab.c
  SHIPPED scripts/kconfig/zconf.lex.c
  SHIPPED scripts/kconfig/zconf.hash.c
  HOSTCC  scripts/kconfig/zconf.tab.o
  HOSTLD  scripts/kconfig/conf
#
# configuration written to .config
```

编译完成后，会生成 image 目录，然后通过 tftp 烧录即可：

```
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode$ ls -la
total 1267328
drwxr-xr-x  7 aaron.feng sigmastar_users      4096 Nov  7 12:22 .
drwxr-xr-x  7 aaron.feng sigmastar_users      4096 Oct 25 17:07 ..
drwxr-xr-x 21 aaron.feng sigmastar_users      4096 Nov  7 12:20 boot
-rw-r--r--  1 aaron.feng sigmastar_users  15800408 Oct 25 17:05 boot-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x  4 aaron.feng sigmastar_users      4096 Nov  7 12:22 images
drwxr-xr-x 26 aaron.feng sigmastar_users      4096 Nov  7 12:22 kernel
-rw-r--r--  1 aaron.feng sigmastar_users 625576086 Oct 25 17:05 kernel-TAKOYAKI-ALPHA005-1024.tar.gz
drwxr-xr-x  8 aaron.feng sigmastar_users      4096 Oct 25 17:04 project
-rw-r--r--  1 aaron.feng sigmastar_users 558616668 Oct 25 17:06 project-TAKOYAKI-ALPHA005-1024.tar.gz
-rwxrw-rw-  1 aaron.feng sigmastar_users      2676 Nov  7 12:04 Release_to_customer.sh
drwxr-xr-x  4 aaron.feng sigmastar_users      4096 Oct 25 17:04 sdk
-rw-r--r--  1 aaron.feng sigmastar_users  97694764 Nov  6 20:49 sdk-TAKOYAKI-ALPHA005-1024.tar.gz
```

默认编译的 SSD_PLAYER 工程不带 wifi 和播放器功能，如需要打开，需按如下方法打开。

全功能 flythings 占用的 flash 空间和内存都比较大，需要 ssd202 & spinand 128M 的配置才可以跑起来，具体配置如下：

1.6. 增大 mma heap 内存配置

Nor flash：

将 project\configs\nvr\i2m\8.2.1\nor.glibc-squashfs.011a.128 如下位置改成 0x3800000：

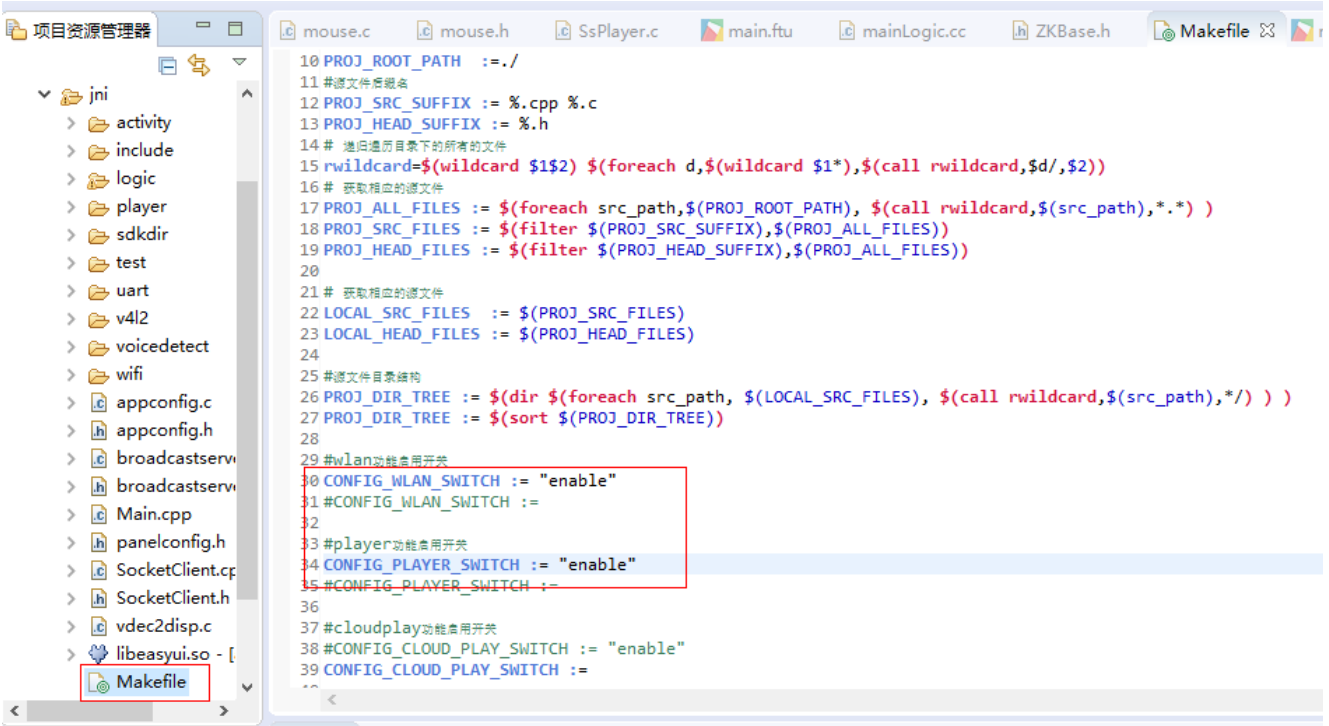
```
CHIP = i2m
BOARD = 011A
BOARD_NAME = SSC011A-S01A
PRODUCT = nvr
TOOLCHAIN = glibc
TOOLCHAIN_VERSION = 8.2.1
KERNEL_VERSION = 4.9.84
LIBC        = libc-2.28
BUSYBOX     = busybox-1.20.2-arm-linux-gnueabihf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = nor.squashfs.hfglibc.nvr.mma
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_128M.h
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x7D00000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070CP50
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLEN) mma_heap=mma_heap_name0,miu=0,sz=0x3800000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabihf-
```

Nand flash:

将 project\configs\nvr\i2m\8.2.1\spinand.glibc.011a.128 的如下位置改成 0x3800000:

```
CHIP = i2m
BOARD = 011A
BOARD_NAME = SSC011A-S01A
PRODUCT = nvr
TOOLCHAIN = glibc
TOOLCHAIN_VERSION = 8.2.1
KERNEL_VERSION = 4.9.84
LIBC        = libc-2.28
BUSYBOX     = busybox-1.20.2-arm-linux-gnueabihf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = spinand.ubifs.hfglibc.p2.mma
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_128M.h
MHAL = i2m
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x7D00000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070BO30I21Y0
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLEN) mma_heap=mma_heap_name0,miu=0,sz=0x3800000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabihf-
```

1.7. 打开 SSD_PLAYER wifi & player 选项：



剩下编译步骤跟精简版 flythings IDE 环境搭建一致。

FLYTHINGS UI 调整触摸屏触控分辨率

触摸屏的触控分辨率需要保持跟 framebuf 的分辨率一致，framebuf 的分辨率参考板子/config/fbdev.ini:

```
/ # cat /config/fbdev.ini
[FB_DEVICE]
FB_HWLAYER_ID = 1
FB_HWWIN_ID = 0
FB_HWLAYER_DST = 3
FB_HWWIN_FORMAT = 5
FB_HWLAYER_OUTPUTCOLOR = 1
FB_WIDTH = 1024
FB_HEIGHT = 600
FB_TIMMING_WIDTH = 1920
FB_TIMMING_HEIGHT = 1080
FB_MMAP_NAME = E_MMAP_ID_FB
FB_BUFFER_LEN = 8192
#unit:Kbyte,4096=4M, fbdev.ko alloc size = FB_BUFFER_L
```

配置方法如下(1024x600.bin & 800x480.bin 放在 ftp：/20191029/SDK/SSD_PLAYER.tar.gz)：

1024x600：

echo 1024x600.bin > /sys/bus/i2c/devices/1-005d/gtcfg

800x480:

echo 800x480.bin > /sys/bus/i2c/devices/1-005d/gtcfg

FLYTHINGS UI 调整 PANEL 分辨率

目前公版 flythings UI 主要适配 1024x600 和 800x480 两种分辨率的 panel,默认的配置是 1024x600 的分辨率，如果要改成 800x480 的分辨率，配置方法如下：

1.8. 屏参修改成 800x480

将 FTP /20191029/SDK/SSD_PLAYER.tar.gz 解压，将 SSD_PLAYER\app\sstardisp.c 的 UI_1024_600 置为 0：

```
#include "string.h"
#include "stdio.h"
#include "stdlib.h"

#include "mi_sys.h"
#include "sstardisp.h"

#include "mi_panel_datatype.h"
#include "mi_panel.h"
#include "mi_disp_datatype.h"
#include "mi_disp.h"


#define UI_1024_600 0
#define USE_MIPI    0


#if UI_1024_600
#include "SAT070CP50_1024x600.h"
#else
#if USE_MIPI
#include "ST7701S_480x480_MIPI.h"
#else
#include "SAT070AT50_800x480.h"
#endif
#endif
```

进入到 SSD_PLAYER\app，重新编译 zkgui，编译前需要将 app 下面的 makefile project 路径改成你实际的 project 路径：

X:\Code\P2\GitHub\Release\SSD_PLAYER\app\Makefile - Notepad++

```
TARGET = zkgui

#赋值对应的路径,此路径需要和
#PROJECT_PATH=/home/beal.wu/i2m_i6_master/project
PROJECT_PATH=/home/aaron.feng/Code/P2/pure_code/release/release_1025/sourcecode/project

MINIGUI_PATH=./minigui_dir

COMPILE_ROOT := $(shell pwd)
CC = arm-linux-gnueabihf-g++
STRIP = arm-linux-gnueabihf-strip
```

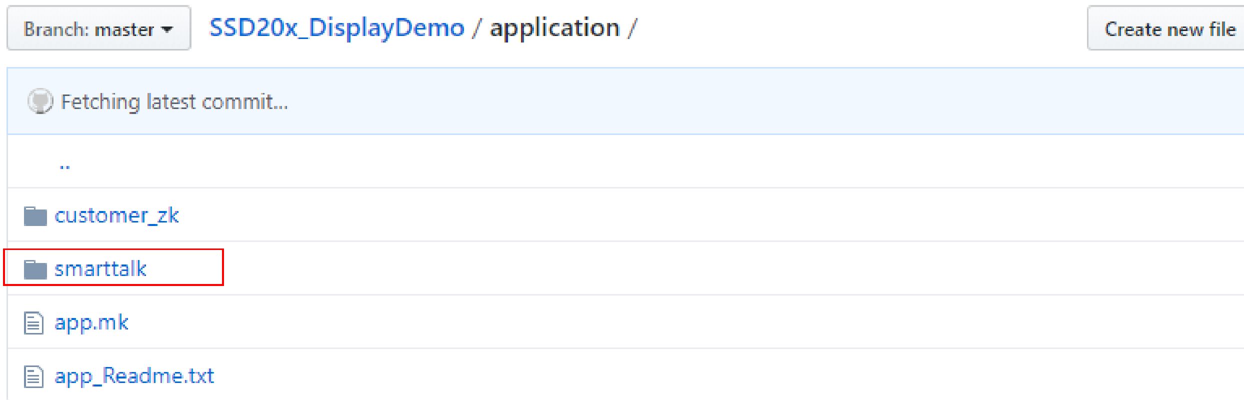
然后直接 make clean;make 即可：

```
aaron.feng@sigmastar:~/Code/P2/GitHub/Release/SSD_PLAYER/app$ make clean;make
rm -rf  Main.cpp.o ./bin/
rm -rf *.o
arm-linux-gnueabihf-g++ -I./minigui_dir/include -I./minigui_dir/include/freetype2 -I./mi
lude -L./minigui_dir/lib -L/home/aaron.feng/Code/P2/pure_code/release/release_1025/sour
hread -lfreetype -ldl -lts -lz -lmi_common -lmi_gfx -lmi_sys -lmi_disp -lmi_panel -lmi_
arm-linux-gnueabihf-g++ -I./minigui_dir/include -I./minigui_dir/include/freetype2 -I./mi
lude -L./minigui_dir/lib -L/home/aaron.feng/Code/P2/pure_code/release/release_1025/sour
hread -lfreetype -ldl -lts -lz -lmi_common -lmi_gfx -lmi_sys -lmi_disp -lmi_panel -lmi_
arm-linux-gnueabihf-g++ -I./minigui_dir/include -I./minigui_dir/include/freetype2 -I./mi
lude -L./minigui_dir/lib -L/home/aaron.feng/Code/P2/pure_code/release/release_1025/sour
hread -lfreetype -ldl -lts -lz -lmi_common -lmi_gfx -lmi_sys -lmi_disp -lmi_panel -lmi_
mkdir -p ./bin
mv zkgui ./bin/
aaron.feng@sigmastar:~/Code/P2/GitHub/Release/SSD_PLAYER/app$
```

编译完将 SSD_PLAYER\app\bin\zkgui 替换到 sdk\verify\application\customer_zk_xx\bin 下面，重名为 zkgui_800_480，然后执行 1.5 步骤，重新编译烧录即可；

2. MINIGUI SMARTTALK 环境搭建

smarttalk 比较大，nor 放不下必须要 spinand 才能跑, demo 放在：
https://github.com/wuyanxiong/SSD20x_DisplayDemo/tree/master/application
将 smarttalk 拷贝到 sdk\verify\application，再按如下步骤配置即可。



2.1. Sdk 打开 smarttalk 编译选项

project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk(smarttalk 和 flythings 不能共存，打开 verify_smarttalk 需要屏蔽 verify_zk_mini):

```
include $(PROJ_ROOT)/release/customer_tailor/nvr_default.mk
#interface_ai:=disable
#interface_ao:=disable
#interface_gfx:=disable
interface_hdmi:=disable
#interface_divp:=disable
#interface_disp:=disable
#interface_panel:=disable
interface_rgn:=disable
interface_shadow:=disable
interface_uac:=disable
interface_vdf:=disable
interface_vdisp:=disable
#interface_vdec:=disable
interface_venc:=enable
interface_wlan:=enable

misc_fbdev:=enable
#verify_zk_mini:=enable
verify_smarttalk:=enable
#mhal
#mhal_aio:=disable
mhal_csi:=disable
#mhal_disp:=disable
```

2.2. 编译 project & sdk & kernel

Smarttalk 播放视频比较耗内存，编译前需要将对应 config 的 mma 的 size 加大到 0x2000000：

*X:\Code\P2\pure_code\release\release_1025\sourcecode\project\configs\nvr\i2m\8.2.1\spinand.glibc.011a.64 - Notepad++

```
CHIP = i2m
BOARD = 011A
BOARD_NAME = SSC011A-S01A
PRODUCT = nvr
TOOLCHAIN = glibc
TOOLCHAIN_VERSION = 8.2.1
KERNEL_VERSION = 4.9.84
LIBC       = libc-2.28
BUSYBOX    = busybox-1.20.2-arm-linux-gnueabihf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = spinand.ubifs.hfglibc.p2.mma
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_64M.h
MHAL = i2m
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x3D000000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070BO30I21Y0
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLEN) mma_heap=mma_heap_name0,miu=0,sz=0x2000000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabihf-
```

编译烧录步骤跟 1.5 一致

2.3. 调整屏幕分辨率

公版目前主要适配了 1024x600 和 800x480 两种分辨率的 panel，默认配置是 800x480，如果需要改成 1024x600 配置方法如下：

2.3.1 修改屏参文件

sdk\verify\application\smarttalk\demoApp\main_ui.h 打开 UI_1024_600:

```
#define UI_ARGB888_FF UI_ARGB2PIXEL8888(255,255,255,255)

#define UI_ARGB888_KEY UI_ARGB2PIXEL8888(128,128,128,128)


#define UI_DISP_W           1920
#define UI_DISP_H           1080

#define UI_1024_600

#ifdef UI_1024_600
#define MAINWND_W           1024
#define MAINWND_H           600
#else
#define MAINWND_W           800
#define MAINWND_H           480
#endif
```

重新 build sdk\verify\application\smarttalk\demoApp：

```
-rwxr--r-- 1 aaron.feng sigmastar_users    489 Nov  8 12:41 utility.h
aaron.feng@sigmastar:~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/verify/application/smarttalk/demoApp$ rm -rf build;./build_sample.sh
~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/verify/application/smarttalk/demoApp/build ~/Code/P2/pure_code/release/release_1025/sourcecode/sdk/ver
-- The C compiler identification is GNU 5.4.0
-- The CXX compiler identification is GNU 5.4.0
-- Check for working C compiler: /usr/bin/cc
-- Check for working C compiler: /usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
```

2.3.2 修改 framebuf 宽高和大小

project\board\i2m\SSC011A-S01A\config\fbdev.ini:

```
[FB_DEVICE]
FB_HWLAYER_ID = 1
FB_HWWIN_ID = 0
FB_HWLAYER_DST = 3
FB_HWWIN_FORMAT = 5
FB_HWLAYER_OUTPUTCOLOR = 1
FB_WIDTH = 1024
FB_HEIGHT = 600
FB_TIMMING_WIDTH = 1920
FB_TIMMING_HEIGHT = 1080
FB_MMAP_NAME = E_MMAP_ID_FB
FB_BUFFER_LEN = 8196
#unit:Kbyte,4096=4M, fbdev.ko alloc size = FB_BUFFER_LEN*1024
```

2.3.3 修改 MiniGUI.cfg 相关配置

sdk\verify\application\smarttalk\RunFile\etc\MiniGUI.cfg

```

[system]
# GAL engine and default options
#gal_engine=fbcon
gal_engine=sstar
#defaultmode=800x480-32bpp
mmaBufSize=0xF00000
defaultmode=1024x600-32bpp

# IAL engine
#ial_engine=console
ial_engine=tslib
#mdev=/dev/input/mice
mdev=/dev/input/event0
mtype=IMPS2
#mtype=none

[sstar]
#defaultmode=800x480-32bpp
defaultmode=1024x600-32bpp
```

2.3.4 修改 mini gui 布局文件

sdk\verify\application\smarttalk\ image.mk:

```
app:
ifeq ($(IMAGE_INSTALL_DIR),)
	@echo "directory of image is not defined"
	@exit 1
endif
	@cp $(APPLICATION_PATH)/smarttalk/demoApp/build/mginit $(IMAGE_INSTALL_DIR)/customer
	@$(TOOLCHAIN_REL)strip --strip-unneeded $(IMAGE_INSTALL_DIR)/customer/mginit
	@cp -rf $(APPLICATION_PATH)/appres  $(IMAGE_INSTALL_DIR)/customer
	@cp -rf $(APPLICATION_PATH)/smarttalk/RunFile/CSpotter  $(IMAGE_INSTALL_DIR)/customer
	@cp -rf $(APPLICATION_PATH)/smarttalk/RunFile/minigui $(IMAGE_INSTALL_DIR)/customer
	@cp -rf $(APPLICATION_PATH)/smarttalk/RunFile/lib $(IMAGE_INSTALL_DIR)/customer
	@cp -rf $(APPLICATION_PATH)/smarttalk/RunFile/etc/* $(IMAGE_INSTALL_DIR)/rootfs/etc/
	@cat $(APPLICATION_PATH)/smarttalk/RunFile/demo.sh >> $(IMAGE_INSTALL_DIR)/customer/demo.sh
	@cp $(APPLICATION_PATH)/smarttalk/RunFile/hscrollview.rc $(IMAGE_INSTALL_DIR)/customer
	@cp $(APPLICATION_PATH)/smarttalk/RunFile/*.xml $(IMAGE_INSTALL_DIR)/customer
	@cp $(APPLICATION_PATH)/smarttalk/RunFile/720P25.h264 $(IMAGE_INSTALL_DIR)/customer
	@cp $(APPLICATION_PATH)/smarttalk/RunFile/8K_16bit_MONO.wav $(IMAGE_INSTALL_DIR)/customer
	#@mv $(IMAGE_INSTALL_DIR)/customer/smartLayout_800x480.xml $(IMAGE_INSTALL_DIR)/customer/smartLayout.xml
	@mv $(IMAGE_INSTALL_DIR)/customer/smartLayout_1024x600.xml $(IMAGE_INSTALL_DIR)/customer/smartLayout.xml
```